

Decision Modelling (NPTEL Online Course 2021)

Assignment 3 (Queuing Theory- L11-L15)

Detailed Answers

1. Queuing models determine the ways to operate a queuing system making a tradeoff between:
 - i. **Cost of waiting and cost of providing service**
 - ii. Cost of arrival and cost of waiting
 - iii. Cost of arrival and cost of providing service
 - iv. Cost of waiting and cost of queue space
2. If the service rate remains constant and the arrival rate decreases, then, in general:
 - i. Customer waiting time increases
 - ii. **Customer waiting time decreases**
 - iii. Service costs increase
 - iv. Customer satisfaction decreases
3. A customer goes to a Banking facility to withdraw money and joins the queue before one ATM. After a while, she finds it is taking more time than expected and hence she leaves the queue and joins the queue in front of another ATM. This the case of:
 - i. Balking
 - ii. Reneging
 - iii. **Jockeying**
 - iv. Alternating
4. A courier delivery boy keeps all the delivery packages in a bag one atop the other and starts delivering sequentially from top. Which kind of queue discipline is illustrated here?
 - i. First-come-first-served (FCFS)
 - ii. Service in random order (SIRO)
 - iii. **Last in first out (LIFO)**
 - iv. First in first out (FIFO)
5. The condition $p(X > (t + \Delta t) \mid X > \Delta t) = p(X > t)$ shows which of the following property of exponential distribution:
 - i. Decreasing property
 - ii. **Memory-less property**
 - iii. Maximum entropy distribution
 - iv. None of the above

6. Which of the following relationships is not true?
- i. $W = W_q + 1/\mu$
 - ii. $L = \lambda W$
 - iii. $L = L_q + \lambda$**
 - iv. $L_q = \lambda W_q$
7. Which among the following is not a property of a Poisson process?
- i. Stationarity
 - ii. Orderliness
 - iii. Independence
 - iv. Linearity**
8. As per Kendall notation, the letter 'c' indicates:
- i. Probability distribution of the service time
 - ii. Queue discipline
 - iii. Number of servers in the system**
 - iv. Probability distribution of the arrivals
9. As per Kendall notation, the letter 'a' indicates:
- i. Probability distribution of the arrivals**
 - ii. Maximum number of customers allowed in the system
 - iii. Probability distribution of the service time
 - iv. Queue discipline
10. The time required to deliver orders at a fast-food pickup point having single operator has an exponential distribution with a mean of 150 seconds. If the customer's arrival follows Poisson distribution, with mean rate 22 per hour, what is the average time the customer waits in the system?
- i. 34 minutes
 - ii. 32 minutes
 - iii. 30 minutes**
 - iv. 28 minutes

Answer to Q10.

Arrival rate, $\lambda = 22$ per hour

Service time = 150 sec = $1/24$ hour

Service rate, $\mu = 24$ per hour

Average waiting time of a customer in the system = $1/(\mu - \lambda) = 1/2$ hours = **30 minutes**

11. In a juice center, the estimated arrival rate of customer is 15 per hour on an average. If it takes 3 minutes on an average to serve the order, what is the average time the customer need to wait to get his order served?
- i. 9 minutes
 - ii. **12 minutes**
 - iii. 15 minutes
 - iv. None of the above

Answer to Q11.

Arrival rate, $\lambda = 15$ per hour

Service time = 3min = $1/20$ hour

Service rate, $\mu = 20$ per hour

Average waiting time of a customer in the system = $1/(\mu - \lambda) = 1/5$ hours = **12 minutes**

12. An AC serviceman finds that the time spent on his job has an exponential distribution with a mean of 18 minutes. If he repairs sets in the order in which they come in, and if the arrival of sets is approximately Poisson with an average rate of 3 per hour. The serviceman works for 8 hours a day. What is the serviceman's expected idle time each day?
- i. 18 minutes
 - ii. 30 minutes
 - iii. **48 minutes**
 - iv. 1 hour

Answer to Q12.

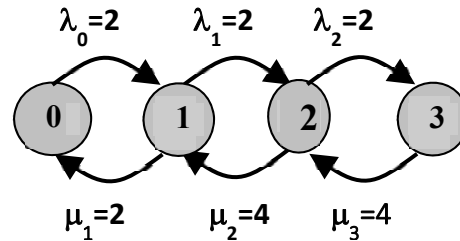
Arrival rate, $\lambda = 3$ per hour

Service rate, $\mu = 60/18 = 3.33$ per hour

Utilization, $\rho = \lambda/\mu = 3/3.33 = 0.90$

Idle time = $(1 - \rho) * 8 * 60 = (1 - 0.9) * 480 = \mathbf{48 \text{ minutes}}$

13. For the Rate diagram shown below, which of the following relationship is true at Node 3:



- i. $P_2 = P_3$
- ii. **$2P_2 = 4P_3$**
- iii. $4P_2 = 2P_3$
- iv. None of these

14. In a vaccination center, only one person can get vaccinated at a time. During the time someone taking the vaccine, any other arriving person has to wait in the waiting area. The people are arriving randomly at 10 per hour. The vaccination process is exponential and takes 5 minutes on the average. What is the probability that an arriving person will have to wait in the waiting area?

- i. 75%
- ii. 78%
- iii. 80%
- iv. **83%**

Answer to Q14.

Average arrival rate $\lambda = 10$ per hour.

Average service time $1/\mu = 5$ minutes = $5/60$ hour = $1/12$ hour, and hence $\mu = 12$ /hour.

Hence, the utilization factor, $\rho = \lambda/\mu = 10/12 = 5/6$

So, the probability that an arrival will find the place free = $p(0) = 1 - \rho = 1 - (5/6) = 1/6$

An arriving customer has to wait outside when there is no place free inside.

So, probability that an arriving customer has to wait outside:

= $1 - \text{probability that the arrival finds the place free}$

= $1 - (1/6) = 5/6 = \mathbf{83\%}$

15. Arrivals to an underground parking facility having single entry are Poisson distributed with a rate of 40 per hour. The average time to cross the entryway to actual parking area is 45 seconds and this time is exponentially distributed. Find the utilization of the entryway.

- i. **50%**
- ii. 60%
- iii. 75%
- iv. 80%

Answer to Q15.

Arrival rate $\lambda = 40$ per hour

Service time = 45 sec = $3/4$ minute

Service rate $\mu = 60/(3/4) = 80$ per hour

Runway utilization, $\rho = \lambda/\mu = 40/80 = 1/2 = 50\%$

16. In the enrollment counter of a college with single data entry operator, there are two chairs meant for students waiting for enrollment. On an average, one student arrives every 15 minutes and it takes 5 minutes for the enrollment process to complete. Find the probability that a student on arrival finds at least one chair free.

- i. 92%
- ii. 94%
- iii. 95%
- iv. 96%**

Answer to Q16.

Average inter-arrival time $1/\lambda = 15$ minutes,

Average service time $1/\mu = 5$ minutes.

Hence, the utilization factor, $\rho = \lambda/\mu = 5/15 = 1/3$

Now, the probability that a student on arrival finds at least one chair free

= probability that there is no one in the system $p(0)$

+ probability that there is only one person in the system and is in service $p(1)$

+ probability that there are two person in the system and one is in service $p(2)$

$$p(0) = 1 - \rho = 1 - 1/3 = 2/3 = 0.67$$

$$p(1) = \rho * p(0) = 1/3 * 2/3 = 2/9 = 0.22$$

$$p(2) = \rho * p(1) = 1/3 * 2/9 = 2/27 = 0.07$$

So, the required probability = $0.67 + 0.22 + 0.07 = 0.96$

17. In a college enrollment counter with a single operator, on the average one student arrives every 15 minutes in the forenoon whereas one student arrives every 10 minutes in the afternoon. If it takes 5 minutes for the enrollment process to complete, which of the following statements correctly describes the average waiting time in the system?

- i. The waiting time in forenoon is 3 min more than that in the afternoon
- ii. The waiting time in forenoon is 3 min less than that in the afternoon
- iii. The waiting time in forenoon is 2.5 min more than that in the afternoon
- iv. The waiting time in forenoon is 2.5 min less than that in the afternoon**

Answer to Q17.

Forenoon enrollment

Average inter-arrival time $1/\lambda = 15$ minutes, so Arrival rate, $\lambda = 1/15$

Service time $= 1/\mu = 5$ minutes; Service rate, $\mu = 1/5$ minutes

Average waiting time of a customer in the system $= 1/(\mu - \lambda) = 1/(1/5 - 1/15) = 15/2 = 7.5$ minutes

Afternoon enrollment

Average inter-arrival time $1/\lambda = 10$ minutes, so Arrival rate $\lambda = 1/10$

Service time $= 1/\mu = 5$ minutes; Service rate $\mu = 1/5$ minutes

Average waiting time of a customer in the system $= 1/(\mu - \lambda) = 1/(1/5 - 1/10) = 10$ minutes

So, Average waiting time is **2.5 minutes less** in the forenoon compared to that in the afternoon.

18. In a small diagnostic center, only one person can give his blood sample at a time. In summer, the arrival rate of patients is 12 per hour whereas in monsoon, it is 18 per hour. The sample collection process is exponential and takes 3 minutes on the average. What is the difference between the average numbers of patients in the system in the two seasons?
- i. 8.5
 - ii. 6.5
 - iii. **7.5**
 - iv. 9.5

Answer to Q18.

In summer

Arrival rate $\lambda = 12$ per hour $= 12/60$ per minute $= 1/5$ per minute

Service time $= 3$ minute; Service rate $\mu = 1/3$ per minute

Average number of patients in the system, $L = \lambda/(\mu - \lambda) = (1/5)/(1/3 - 1/5) = 1.5$

In monsoon

Arrival rate $\lambda = 18$ per hour $= 18/60$ per minute $= 3/10$ per minute

Service time $= 3$ minute; Service rate $\mu = 1/3$ per minute

Average number of patients in the system, $L = \lambda/(\mu - \lambda) = (3/10)/(1/3 - 3/10) = 9$

Difference in the number of patients in the system in the two season is $9 - 1.5 = 7.5$

19. A ticket booking counter has arrival of customer Poisson distributed with a rate of 45 per hour. The average time to book a ticket is 20 seconds and this time is exponentially distributed. Due to some problem the service time increases to 30 seconds. Find the change in utilization of the booking counter.

- i. **12.5% increased**
- ii. 12.5% decreased
- iii. 10.5% increased
- iv. 10.5% decreased

Answer to Q19.

Before the problem appeared

Arrival rate $\lambda = 45$ per hour $= 45/60$ per minute $= 3/4$ per minute

Service time $= 20$ sec $= 1/3$ minute; Service rate $\mu = 3$ per minute

Counter utilization, $\rho = \lambda / \mu = (3/4)/(3) = 1/4 = 25 \%$

After the problem appeared

Arrival rate $\lambda = 45$ per hour $= 45/60$ per minute $= 3/4$ per minute

Service time $= 30$ sec $= 1/2$ minute; Service rate $\mu = 2$ per minute

Counter utilization, $\rho = \lambda / \mu = (3/4)/(2) = 3/8 = 37.5 \%$

Change in utilization is $(37.5 - 25) = 12.5\%$

20. In a city there are three fast-food delivery point of online orders of a popular chain of restaurant named alpha, beta and gamma. A customer having the following information as shown in the table want to have food in the fastest possible way. What should be his preferred choice of delivery point?

	Alpha	Beta	Gamma
Order arrival rate per hour	18	20	21
Order delivery time	2 min	2min 15 sec	2 min
Travel time from his location	10 min	8 min	12 min

- i. **Alpha**
- ii. Beta
- iii. Gamma
- iv. Cannot be decided

Answer to Q20.

	Alpha	Beta	Gamma
Order arrival rate per hour (λ)	18 per hour = 3/10 per min	20 per hour = 1/3 per min	21 per hour = 7/20 per min
Order delivery time	2 min	2min 15 sec = 9/4 min	2 min
Service rate (μ)	1/2 per min	4/9 per min	1/2 per min
Waiting time , $W = 1/(\mu - \lambda)$	$1/(1/2 - 3/10)$ = 10/2 = 5 min	$1/(4/9 - 1/3)$ = 9 min	$1/(1/2 - 7/20)$ = 20/3 = 6 min 40 sec
Travel time from his location, T	10 min	8 min	12 min
Total time taken = W+T	15 min	17 min	18 min 40 sec

Hence preferred delivery point choice would be **Alpha**.